

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 11-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-16** **Max. Marks: 180**

11-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P2)_GTA-16_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 9 – 16)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Let $f(x)=\lim\limits_{n\to\infty}\tan^{-1}\left(4n^2\left(1-\cos\frac{x}{n}\right)\right)$ and $g(x)=\lim\limits_{n\to\infty}\frac{n^2}{2}\ln\cos\left(\frac{2x}{n}\right)$ then $\lim\limits_{x\to0}\frac{e^{-2g(x)}-e^{f(x)}}{x^6}$ equals

A) $\frac{8}{3}$ **B)** $\frac{7}{3}$ **C)** $\frac{5}{3}$ **D)** $\frac{2}{3}$

2. The area enclosed between the curves $|x|+|y|\geq 2$ and $y^2=4\left(1-\frac{x^2}{9}\right)$ is :

A) $(6\pi-4)$ sq. units **B)** $(6\pi-8)$ sq. units
C) $(3\pi-4)$ sq. units **D)** $(3\pi-2)$ sq. units

3. The director circle of the parabola $(y-2)^2=16(x+7)$ touches the circle $(x-1)^2+(y+1)^2=r^2$, then 'r' is equal to

A) 10 **B)** 11 **C)** 12 **D)** 16

4. In the xy-plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x-6)^2+(y-8)^2=25$ is :

A) $10\sqrt{3}$ **B)** $10\sqrt{5}$ **C)** $10\sqrt{3}+\frac{5\pi}{3}$ **D)** $10+5\pi$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. An ellipse has semi-major axis of length 2 and semi-minor axis of length 1. It slides between the co-ordinate axes in the first quadrant, while maintaining contact with both x-axis and y-axis

A) The locus of the centre of ellipse is $x^2+y^2=3$

B) The locus of the centre of ellipse is $x^2+y^2=5$

C) The locus of the foci of the ellipse is $x^2+y^2+\frac{1}{x^2}+\frac{1}{y^2}=16$

D) The locus of the foci of the ellipse is $x^2+y^2+\frac{1}{x^2}-\frac{1}{y^2}=2\sqrt{3}+4$

6. If $\alpha=\max\left\{(x+2)^2+(y-3)^2\right\}$ and $b=\min\left\{(x+2)^2+(y-3)^2\right\}$ where x, y satisfying $x^2+y^2+8x-10y-40=0$, then :

A) a + b = 18 **B)** a + b = 178 **C)** $a-b=4\sqrt{2}$ **D)** $a-b=72\sqrt{2}$

7. Which of the following is/are true for the function $f(x)=\int_0^x \frac{\cos t}{t}\,dt\,(x>0)$?

A) f(x) is monotonically increasing in $\left((4n-1)\frac{\pi}{2},(4n+1)\frac{\pi}{2}\right)\forall n\in N$

B) f(x) has a local minima at $x=(4n-1)\frac{\pi}{2}\forall n\in N$

C) The points of inflection of the curve y = f(x) lie on the curve xtanx + 1 = 0

D) Number of critical points of y = f(x) in $(0,10\pi)$ are 19

8. If the numerical value of $\tan\left(\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right)$ is $\left(\frac{a}{b}\right)$, where a, b are two positive integers and their H.C.F is 1

A) a + b = 23 **B)** a – b = 11 **C)** 3b = a + 1 **D)** 2a = 3b

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SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

9. For a unique value of μ and λ, the system of equations given by

$x + y + z = 6$

$x + 2y + 3z = 14$

$2x + 5y + \lambda z = \mu$

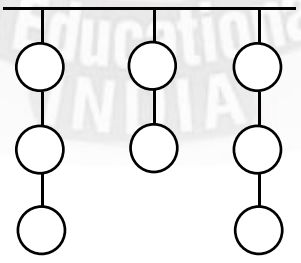
Has infinitely may solutions, then $\frac{\mu - \lambda}{4}$ is equal to :

10. For any increasing sequence of real numbers $a_1, a_2, a_3, \ldots\ldots$ let us denote A as $<a_1, a_2, a_3, a_4, \ldots\ldots>$ and a sequence ΔA is defined as $< a_2 - a_1, a_3 - a_2, a_4 - a_3, \ldots\ldots>$ Suppose that all of the terms of the sequence $\Delta(\Delta A)$ are 1 and that $a_{19} = a_{92} = 0$. Then the sum of digit(s) of a_3 is M, then the value of $\left[\frac{M}{2}\right]$ is ….. , where [.] is greatest integer function.

11. Let $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$. Function f(x) satisfies $\int_0^2 f(x)\,dx = 5$.

If $\int_0^{50} f(x)\,dx = I$. Find $\left[\sqrt{I} - 3\right]$. (Where [.] denotes greatest integer function)

12. 8 clay targets are arranged as shown. If 'N' be the number of different ways they can be shot (one at a time) if no target can be shot until the target(s) below it have been shot. Find the ten's digit of N.



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13. Let $a = 3^{\frac{1}{223}} + 1$ and for all $n \geq 3$, let

$f(n) = {}^{n}C_0 . a^{n-1} - {}^{n}C_1 . a^{n-2} + {}^{n}C_2 . a^{n-3} - \ldots\ldots\ldots + (-1)^{n-1} {}^{n}C_{n-1} . a^0$

If the value of $f(2007) + f(2008) = 3^7 k$ where $k \in N$ then find 'k'

14. Let complex number 'z' satisfy the inequality $2 \leq |z| \leq 4$. A point 'P' is selected in this region at random. The probability that argument of 'P' lies in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is $\frac{1}{K}$, then K =

15. Give that for a, b, c, d $\in$ R, if

$a\sec(200^0) - c\tan(200^0) = d$ and $b\sec(200^0) + d\tan(200^0) = c$, then find the value of

$\left(\frac{a^2 + b^2 + c^2 + d^2}{bd - ac}\right)\sin 20^0$.

16. Let $f(x) = -4.e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ and 'g' be inverse function of 'f' and

$h(x) = \frac{a + bx^{3/2}}{x^{5/4}}$, $h'(5) = 0$, then $\frac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} =$

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PHYSICS **Max Marks: 60**

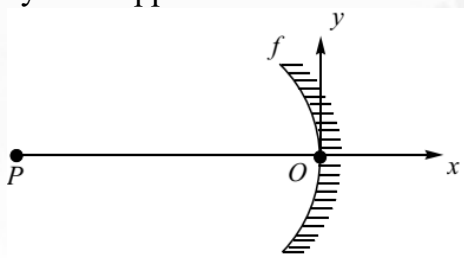
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

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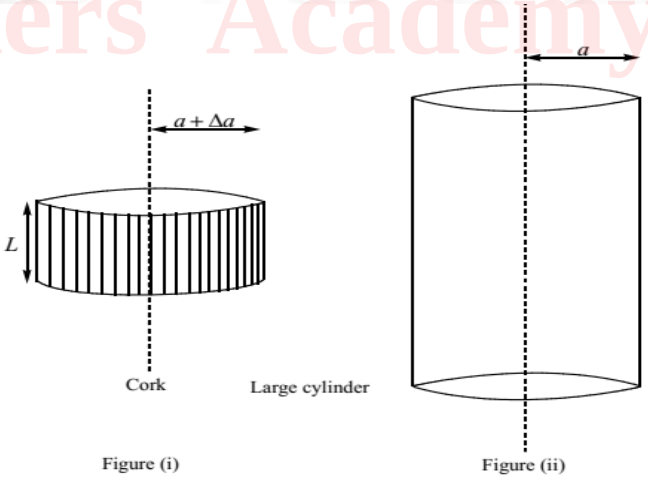
17. A concave mirror has a focal length f = 15 cm. A point object is kept at its centre of curvature. Now mirror is rotated about z-axis through a very small angle $\theta = 2^0$. Consider paraxial rays only. The approximate distance between object and its image is



A) $\frac{\pi}{150}$m **B)** $\frac{\pi}{75}$m **C)** $\frac{\pi}{300}$m **D)** $\frac{\pi}{100}$m

18. A large steel cylinder has inner radius 'a' as shown in the figure-(ii). There is a cork of radius $a + \Delta a(\Delta a << a)$ and length 'L' shown in the figure (i). Now, cork is to be fitted to close the open mouth of cylinder. Consider poisson's ratio of cork is almost zero and its bulk modulus is 'B'.

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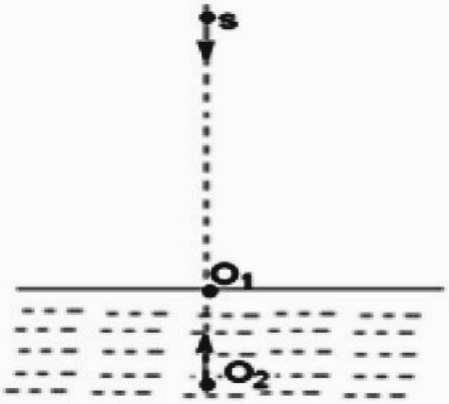


If μ is the coefficient of friction between cork and container then the needed force to pull it out of the container is

A) $2\pi\mu BL(\Delta a)$ **B)** $8\pi\mu BL(\Delta a)$ **C)** $4\pi\mu BL(\Delta a)$ **D)** $\pi\mu BL(\Delta a)$

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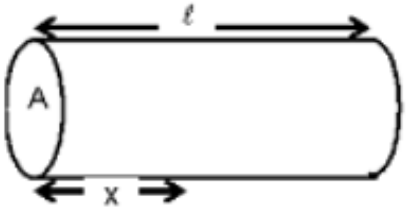
19. In the figure shown an observer O_1 floats (at rest) on water surface while another observer O_2 is moving upwards with constant velocity $v_1 = \frac{v}{5}$ in water. The source (s) moves down with constant velocity $v_s = \frac{v}{5}$ and emits sound of frequency f. The velocity of sound in air is v and that in water is 4v. Then,



the frequency of the sound received by O_2 is

A) $\frac{21f}{8}$ **B)** $\frac{7f}{4}$ **C)** $\frac{16f}{21}$ **D)** $\frac{21f}{16}$

20. The cross-sectional area and length of cylindrical conductor are A and l respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 l / \sqrt{x}$ where 'x' is the distance along the axis of the cylinder from one of its ends. A potential difference V_0 is applied across the cylinder. Choose the incorrect statement(s)



A) The resistance of the system along the cylindrical axis is $\frac{2\sqrt{l}}{3A\sigma_0}$

B) The current density is constant throughout the length of the cylinder.

C) The electric field E(x) at each point in the cylinder is $E(x) = \frac{5V_0\sqrt{x}}{2l^{3/2}}$

D) The electric field E(x) at each point in the cylinder is $E(x) = \frac{3V_0\sqrt{x}}{2l^{3/2}}$

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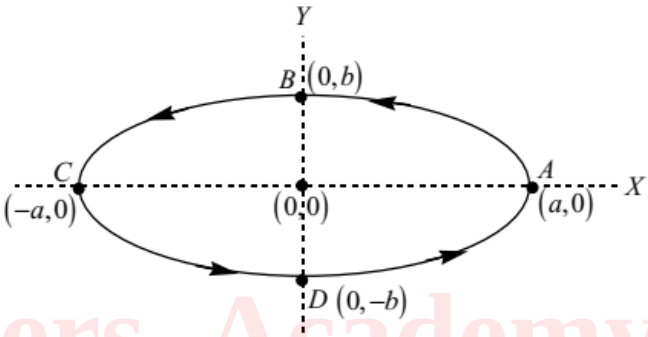
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions

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21. A particle of mass 'm' is taken with a constant speed V_0 along the circumference of an elliptical orbit with a semi major axis of length '*a*' and semi minor axis of length 'b' as shown in fig. then which of the following statements are correct :



A) Radius of curvature of the elliptical orbit at (-*a*, 0) is $\frac{b^2}{a}$

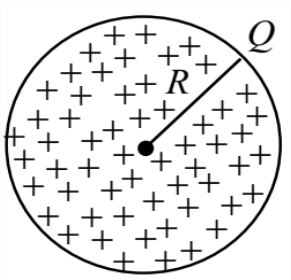
B) Radius of curvature of the elliptical orbit at (0, -b) is $\sqrt{\frac{a^2+b^2}{2}}$

C) The magnitude of angular momentum of the particle when it crosses the (0, b) about the origin is mV_0b

D) The magnitude of acceleration of the particle when it crosses point (0, -b) is $\frac{V_0^2 b}{a^2}$

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22. A Non-conducting solid sphere of radius 'R' and charge '+Q' is distributed uniformly throughout the volume, as shown in fig. (Relative permittivity inside and out side the Non-conducting sphere is taken as one) then which of the following statements are correct



A) The electro-static potential energy in the form of the electric field with in the Non conducting solid sphere is $\frac{Q^2}{40\pi\varepsilon_0 R}$

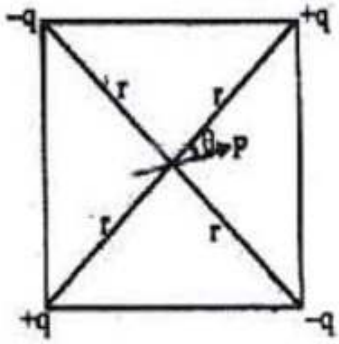
B) The electro –static potential energy in the form of the electric field out side the non conducting solid sphere (from the surface to infinity) is $\frac{Q^2}{8\pi\varepsilon_0 R}$

C) The self energy of the non-conducing charged solid sphere (neglect gravitational self energy) $\frac{3Q^2}{20\pi\varepsilon_0 R}$

D) Electro static potential of the non-conducting sphere at it's centre (infinity is taken as zero reference for the electric potential) is $\frac{3Q^2}{8\pi\varepsilon_0 R}$

23. Given square frame of diagonal length 2r made of insulating wires. There is a short ideal electric dipole, having dipole moment P, fixed in the plane of the figure lying at the centre of the square, making an angle θ as shown in figure. Four identical particles having charges of magnitude 'q' each and alternatively positive and negative are placed at the four corners of the square. Select the correct alternative(s). (neglect gravitational effects) $\left(k = \frac{1}{4\pi\varepsilon_0}\right)$ (Centre of dipole coincidence with the intersecting point of diagonals)

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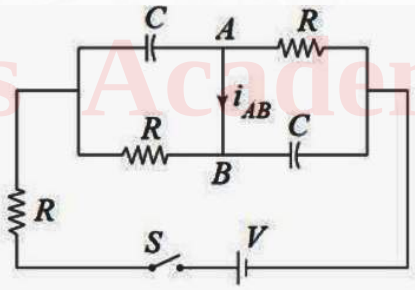
A) Resultant electrostatic force on the system of four charges due to dipole is $\frac{6kPq}{r^3}$

B) Resultant electrostatic force on the system of four charges due to dipole is $\frac{6kPq}{r^3}\cos\theta$

C) Net torque on the system of four charges about the centre of the square due to dipole is zero

D) Net torque on the system of four charges about the centre of the square due to dipole is $\frac{3kPq}{r^2}$

24. In the circuit shown capacitors are uncharged initially. S is a switch that is closed at time $t = 0$. Which of the following statements is/are true?



A) As soon as the switch is closed $i_{AB} = 0$

B) At steady state the power dissipated in the circuit is $\frac{V^2}{3R}$

C) $i_{AB} = \frac{4V}{3R}\left(e^{-\frac{3t}{RC}} - \frac{1}{4}\right)$

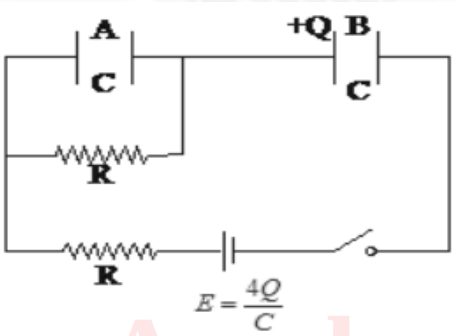
D) $i_{AB} = \frac{V}{3R}\left(e^{-\frac{3t}{4RC}} - 1\right)$

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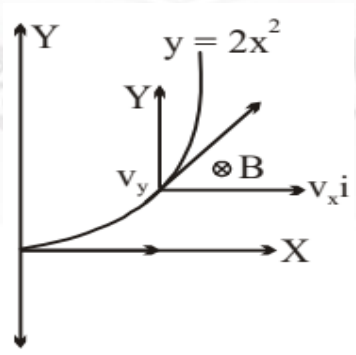
SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

25. Consider the circuit shown in the figure. Capacitors A and B, each have capacitance C = 2F. The plates of capacitor A are shorted using a wire of resistance R = 1Ω while the left plate of capacitor B is given an initial charge Q = +4C. The switch is closed at a time t = 0. What will be the initial current (in ampere) drawn from the battery immediately after the switch is closed?



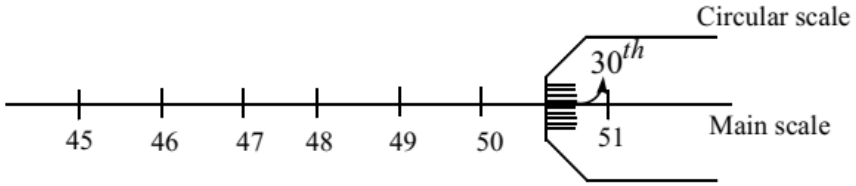
26. A non-uniform magnetic field B varies with 'x'. It exist for $x \geq 0$, into the xy plane. A charge 'q' of mass 'm', enters the magnetic field at the origin with speed $v\hat{i}$. It is seen that it travels along $y = 2x^2$ curve. Find the value of B at x = 0. $\left(\text{given}: \frac{mv}{q} = 2\right)$



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27. In a nuclear reactor an element X decays to a radio active element Y at a constant rate 10^{15} atoms per sec. Each decay releases 100 *MeV* energy. Half life of Y equals T and decays to a stable product Z. Each decay of Y releases 50 *MeV*. All energy released inside the reactor is used to produce electricity at an efficiency of 30%. Calculate the electrical power in *kW* generated in the reactor in steady state.

28. The diameter of a thin wire is measured by using a screw gauge having 100 divisions on circular scale. When spindle of screw gauge is touched to anvil, then zero of circular scale lies 6 division above the reference line of main scale. Now, when wire is kept between spindle and anvil then 30th division of circular scale exactly coincides with the reference line of main scale as shown in the figure. The distance between two successive divisions of main scale is 0.1 mm. The measured diameter of wire is found to be *5.XYZ mm* ,where X, Y and Z are non negative single digit integers. Find the value of $X + Y + Z$ (pitch of the screw gauge is 0.1mm)



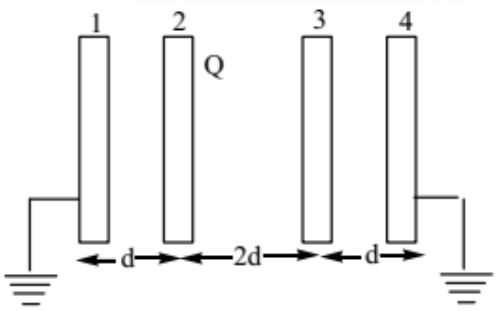
29. Consider a cylinder of height H and radius R. The side wall of container has very small holes which are uniformly distributed to its entire surface area. The number of holes are very large and occupying a fraction k = 0.003 of the entire area of the wall. Water is being poured to container at a constant rate Q. To maintain the water level at height H/2, the value of Q is $\frac{\pi}{100\lambda}R\sqrt{g}H^{3/2}$. Find the value of λ. (Assume torricelli's law is valid)

30. When an $\alpha-$particle with kinetic energy k = 14 MeV collides with a stationary lithium nucleus, the following nuclear reaction takes place. ${}^{4}He + {}^{7}Li \rightarrow {}^{10}B + n$

The *Q*- value of this reaction is E = –2.9 MeV and the neutron is outgoing at right angle to the incoming direction of the $\alpha-$ particle. If kinetic energy of the neutron and the boron nucleus are K_1MeV and K_2MeV respectively. The value of $\frac{K_1}{K_2}$ is $\frac{10n}{61}$. Find 'n'

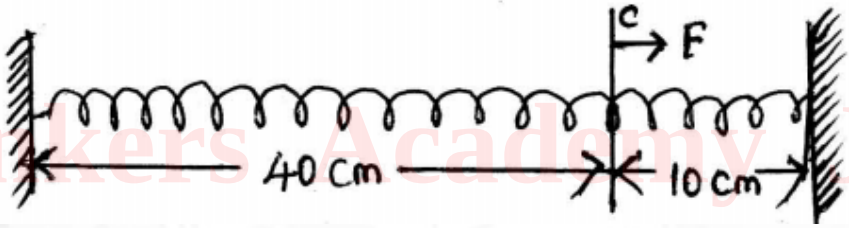
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31. Four metallic plates are placed as shown in the fig. Plate 2 is given a charge 'Q' whereas all other plates are uncharged. Plates 1 and 4 are earthed. The area of each plate is same



The charge appearing on the right side of plate 3 is $\frac{+Q}{k}$. Find 'k'

32. A light uniform spring of force constant 1N/cm is connected between two walls so that the spring is just taut. Point 'C' is located on the spring at distance 40 cm from the left end. Now a force of F = 5N is exerted along the spring at point 'C' as shown. Find the displacement of point 'C' in 'mm' from initial state to final equilibrium state.



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CHEMISTRY **Max Marks: 60**

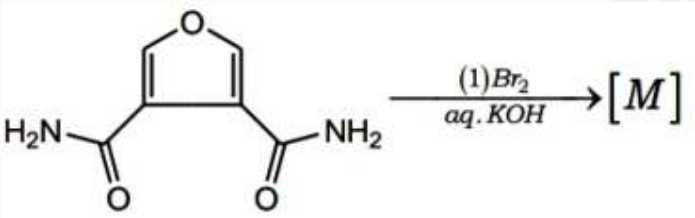
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

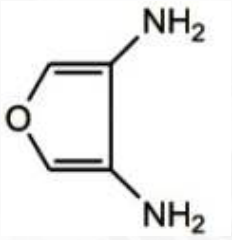
This section contains **FOUR (04)** questions.

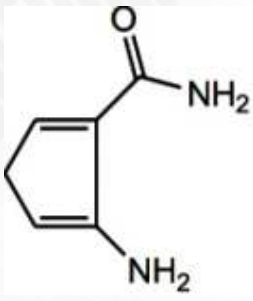
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

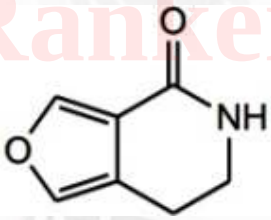
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. Identify the most probable product (M) in the following reaction



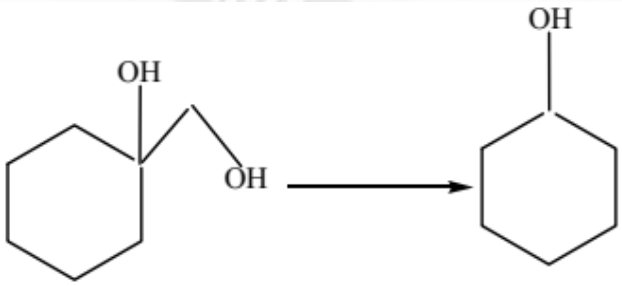
A)


B)


C)


D)


34.



This conversion can be carried out by

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A) $H_2SO_4 / \Delta, NaBH_4$ **B)** $NaIO_4, H^+ / \Delta$

C) $HIO_4, NaBH_4$ **D)** $H^+ / \Delta, Zn(Hg).HCl$

35. Choose the incorrect statement.

A) Ti^{+2} (aq) can liberate hydrogen gas from 1M dilute acid

B) $E^0_{Sc^{+3}/Sc}$ is low due to stability of Sc^{3+}

C) $E^0_{Ni^{+2}/Ni}$ is more negative than expected value due to low ionization energy.

D) Low value for $E^0_{Fe^{+3}/Fe^{+2}}$ attributed to extra stability of Fe^{+3}

36. Initial concentration of the reactant is 1.0M. The concentration becomes 0.9M, 0.8M and 0.7M in 2 hours, 4 hours and 6 hours respectively. Then the order of reaction is

A) 2 **B)** 1 **C)** Zero **D)** 3

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

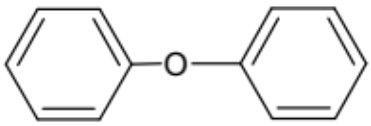
This section contains **THERE (04)** questions.

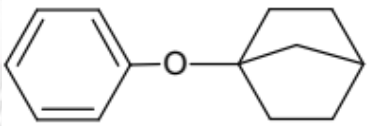
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

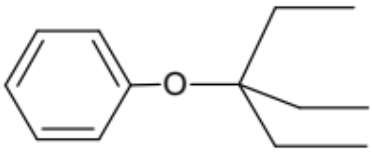
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

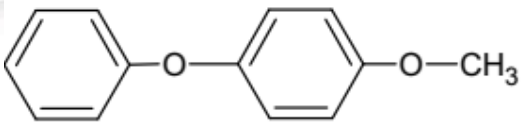
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37. In which of the following ethers C-O bond can be cleaved by conc. HI at high temperature?

A) 

B) 

C) 

D) 

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38. Which of the following statement(s) is/are correct?

A) When BCl_3 gas is passed into an aq. solution of Na_2SO_4, Boric acid is formed

B) Reaction $Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + Cl_2 + 4H_2O$ involves both acid – base reaction and redox reaction

C) Alkali metal hydroxides are more corrosive than alkaline earth metal hydroxides

D) Formation of $Mg^{2+}_{(g)}$ from Mg(g) and $O^{2-}_{(g)}$ from O(g) are endothermic but the formation of MgO(s) from $Mg^{2+}_{(g)}$ and $O^{2-}_{(g)}$ is exothermic

39. Which of the following statement(s) is/are incorrect?

A) Among $\left[Cr(H_2O)_6\right]^{3+}$, $\left[Fe(H_2O)_6\right]^{2+}$ and $\left[Zn(H_2O)_6\right]^{2+}$; $\left[Cr(H_2O)_6\right]^{3+}$ has the highest spin-only magnetic moment value

B) $\left[Ni(CN)_4\right]^{2-}$, $\left[NiCl_4\right]^{2-}$ and $\left[Ni(CO)_4\right]$ are all paramagnetic

C) $\left[PtCl_2(en)_2\right]^{2+}$ has only two stereo isomers

D) $\left[RhCl(Ph_3P)_3\right]$ is known as Wilkinson catalyst and is used for the hydrogenation of alkenes

40. An unknown substance of formula $M^{+2}x^{-2}$, $(M.W = 144)$ crystallizes in one of the cubic lattice, with the density $4.63\ g/cm^3$, then the correct options are (edge length of its unit cell $= 5.87\ \overset{0}{A}$)

A) $M^{+2}x^{-2}$ crystallizes in fcc manner

B) Schottky defect $= 6\times10^{-2}\ gram/cm^3$

C) its unit cell consists 8 ions per unit cell

D) Coordination number of $M^{+2}x^{-2} = 8:8$

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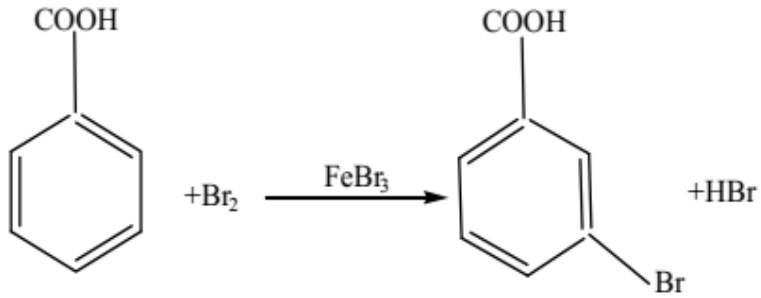
SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
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41. In duma's method for estimation of nitrogen 0.25g of an organic compound gave 44 ml of nitrogen collected at 300K temperature and 725 mm pressure. If the aqueous tension at 300K is 25mm, the percentage of nitrogen in the compound is 'x'. What is the $\frac{x}{2}$ …….

(Round off to the nearest integer)

42.



Consider the above reaction where 6.1 g of benzoic acid is used to get 7.8 g of m-bromo benzoic acid. The percentage yield of the product is ……

Rounded off to the nearest integer

(Given atomic masses C = 12, H = 1.0, O = 16.0, Br=80)

43. The difference in number of singly occupied molecular orbitals (S.O.M.O.) in B_2 and O_2^{2-} is, …….

44. In which of the following reactions paramagnetic gas will form as one of the product.

I) $Ag_2O_{(s)} \xrightarrow{\Delta}$ II) $KClO_{3(s)} \xrightarrow[MnO_2]{\Delta}$

III) $Pb_3O_{4(s)} \xrightarrow{\Delta}$ IV) $Pt + HNO_3 + 3HCl \rightarrow \text{products} + 3(Cl)$

V) $KMnO_4 \xrightarrow{\Delta}$ VI) $NaNO_{2(s)} + \text{dil. } H_2SO_4 \xrightarrow{\Delta}$

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45. To 1 litre solution containing 0.2M Al^{3+} and 0.2M Zn^{2+}, NaOH is added slowly, if concentration of $\left[Al^{3+}\right]$ is 16×10^{-x} at the time when $Zn(OH)_2$ is just beings to precipitate, then find the value of 'x'

$K_{sp}\left[Al(OH)_3\right]=16\times10^{-25}; K_{sp}\left[Zn(OH)_2\right]=2\times10^{-13}$

46. For the electrochemical cell,

$Mg(s)\left|Mg^{2+}(aq.\,1M)\right\|Cu^{2+}(aq.\,1M)\left|Cu(s)\right.$

the standard emf of the cell is 2.70 V at 300 K. When the concentration of Mg^{2+} is changed to ***x***M, the cell potential changes to 2.67 V at 300 K. The value of ***x*** is ……..

(given, $\frac{F}{R}=11500\ KV^{-1}$, where *F* is the Faraday constant and *R* is the gas constant, ln(10) = 2.30)

47. 1.60g of a metal was dissolved in HNO_3 to prepare its nitrate. The nitrate on strong heating gives 2g oxide. The equivalent weight of metal is 4x, then 'x' is

48. Vapour pressure of an equimolar mixture of benzene and toluene at a given temperature was found to be 80mm Hg. If vapour above the liquid phase is condensed in a beaker, vapour pressure of this condensate at the same temperature was found to be 100mm Hg. If the pure state vapour pressure of benzene and toluene is respectively x and y. Then determine value of $\frac{x+2y}{50}$

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31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

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ALL INDIA OPEN CATEGORY RANKS



BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

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